

Experiment 2: For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm in python to output a description of the set of all hypotheses consistent with the training examples.

a)

CGPA	Aptitude Score	Communication Score	Placement
9.2	90	88	Placed
8.8	85	82	Placed
8.5	80	79	Placed
7.2	72	70	Not Placed
6.8	68	65	Not Placed
9	92	90	Placed
7.5	75	74	Not Placed
8.6	84	81	Placed
6.5	60	58	Not Placed
8.9	88	86	Placed

PROGRAM:

```
import pandas as pd
df = pd.read_excel("Dataset.xlsx", sheet_name="Dataset_1", nrows=10)
df["CGPA"] = df["CGPA"].apply(lambda x: "High" if x >= 8 else "Low")
df["Aptitude Score"] = df["Aptitude Score"].apply(lambda x: "High" if x >= 80 else "Low")
df["Communication Score"] = df["Communication Score"].apply(lambda x: "High" if x >= 80 else "Low")
concepts = df.iloc[:, :-1].values.tolist()
target = df.iloc[:, -1].tolist()
def candidate_elimination(concepts, target):
    S = concepts[0].copy()
    G = [['?' for i in range(len(S))] for i in range(len(S))]
    for i, h in enumerate(concepts):
        if target[i] == "Placed":
            for j in range(len(S)):
                if S[j] != h[j]:
                    S[j] = '?'
                    G[j][j] = '?'
        else:
            for j in range(len(S)):
                if h[j] != S[j]:
                    G[j][j] = S[j]
            else:
                G[j][j] = '?'
```

```

G = [g for g in G if g != ['?'] * len(S)]
print("\nSpecific Hypothesis")
print(S)
print("\nGeneral Hypothesis")
for g in G:
    print(g)
candidate_elimination(concepts, target)

```

OUTPUT:

```

Specific Hypothesis
['High', 'High', '?']

General Hypothesis
['High', '?', '?']
['?', 'High', '?']

```

b)

Income (Lakhs)	Credit Score	Loan Amount (Lakhs)	Loan Approved
8	780	5	Yes
7	760	4	Yes
6	720	6	Yes
4	620	8	No
3	580	10	No
9	800	4	Yes
5	650	7	No
8	770	5	Yes
4	600	9	No
7	740	6	Yes

PROGRAM:

```

import pandas as pd
df = pd.read_excel("Dataset.xlsx", sheet_name="Dataset_2", nrows=10)
df["Income (Lakhs)"] = df["Income (Lakhs)"].apply(lambda x: "High" if x >= 6 else "Low")
df["Credit Score"] = df["Credit Score"].apply(lambda x: "Good" if x >= 700 else "Poor")

```

```

df["Loan Amount (Lakhs)"] = df["Loan Amount (Lakhs)"].apply(lambda x: "Small" if x <= 6
else "Large")
concepts = df.iloc[:, :-1].values.tolist()
target = df.iloc[:, -1].tolist()
def candidate_elimination(concepts,target):
    S=concepts[0].copy()
    G=[['?' for i in range(len(S))] for j in range(len(S))]
    for i,h in enumerate(concepts):
        if target[i]=="Yes":
            for j in range(len(S)):
                if S[j]!=h[j]:
                    S[j]='?'
                    G[j][j]='?'
        else:
            for j in range(len(S)):
                if h[j]!=S[j]:
                    G[j][j]=S[j]
            else:
                G[j][j]='?'

    G=[g for g in G if g!=['?']*len(S)]
    print("Specific:",S)
    print("General:")
    for g in G:
        print(g)
candidate_elimination(concepts,target)

```

OUTPUT:

```

Specific: ['High', 'Good', 'Small']
General:
['High', '?', '?']
['?', 'Good', '?']
['?', '?', 'Small']

```

c)

Temperature (°C)	Heart Rate (bpm)	Oxygen Level (%)	Disease
39	110	94	Positive
38.5	108	95	Positive

37	78	99	Negative
39.2	115	93	Positive
36.8	75	98	Negative
38.8	112	94	Positive
37.2	80	98	Negative
39.1	114	92	Positive
36.7	74	99	Negative
38.9	111	93	Positive

PROGRAM:

```

import pandas as pd
df = pd.read_excel("Dataset.xlsx", sheet_name="Dataset_3", nrows=10)
df["Temperature (°C)"] = df["Temperature (°C)"].apply(lambda x: "High" if x >= 38 else
"Normal")
df["Heart Rate (bpm)"] = df["Heart Rate (bpm)"].apply(lambda x: "High" if x >= 100 else
"Normal")
df["Oxygen Level (%)"] = df["Oxygen Level (%)"].apply(lambda x: "Low" if x < 96 else
"Normal")
concepts=df.iloc[:, :-1].values.tolist()
target=df.iloc[:, -1].tolist()
def candidate_elimination(concepts,target):
    S=concepts[0].copy()
    G=[['?' for i in range(len(S))] for j in range(len(S))]
    for i,h in enumerate(concepts):
        if target[i]=="Positive":
            for j in range(len(S)):
                if S[j]!=h[j]:
                    S[j]='?'
                    G[j][j]='?'
        else:
            for j in range(len(S)):
                if h[j]!=S[j]:
                    G[j][j]=S[j]
            else:
                G[j][j]='?'
    G=[g for g in G if g!=['?']*len(S)]
    print("Specific:",S)
    print("General:")
    for g in G:
        print(g)
candidate_elimination(concepts,target)

```

OUTPUT:

```
Specific: ['High', 'High', 'Low']
General:
['High', '?', '?']
['?', 'High', '?']
['?', '?', 'Low']
```

d)

Experience	Performance Score	Training Hours	Promotion
10	95	50	Yes
8	90	45	Yes
7	88	40	Yes
3	65	20	No
2	60	18	No
9	93	48	Yes
4	70	25	No
8	89	42	Yes
2	58	15	No
6	85	38	Yes

PROGRAM:

```
import pandas as pd
df=pd.read_excel("Dataset.xlsx",sheet_name="Dataset_4",nrows=10)
df["Experience"]=df["Experience"].apply(lambda x:"High" if x>=6 else "Low")
df["Performance Score"]=df["Performance Score"].apply(lambda x:"Good" if x>=80 else
"Poor")
df["Training Hours"]=df["Training Hours"].apply(lambda x:"High" if x>=35 else "Low")
concepts=df.iloc[:,:-1].values.tolist()
target=df.iloc[:,:-1].tolist()
def candidate_elimination(concepts,target):
    S=concepts[0].copy()
    G=[['?' for i in range(len(S))] for j in range(len(S))]
    for i,h in enumerate(concepts):
        if target[i]=="Yes":
            for j in range(len(S)):
                if S[j]!=h[j]:
                    S[j]='?'
                    G[j][j]='?'
        else:
```

```

for j in range(len(S)):
    if h[j]!=S[j]:
        G[j][j]=S[j]
    else:
        G[j][j]='?'

G=[g for g in G if g!=['?']*len(S)]
print("Specific:",S)
print("General:")
for g in G:
    print(g)
candidate_elimination(concepts,target)

```

OUTPUT:

```

Specific: ['High', 'Good', 'High']
General:
['High', '?', '?']
['?', 'Good', '?']
['?', '?', 'High']

```

e)

Weight	Diameter	Sweetness	Fruit
150	7.5	90	Apple
160	7.8	88	Apple
140	7.2	91	Apple
120	5.5	70	Orange
125	5.8	72	Orange
130	6	74	Orange
155	7.6	89	Apple
118	5.4	69	Orange
148	7.4	92	Apple
122	5.7	71	Orange

PROGRAM:

```

import pandas as pd

df=pd.read_excel("Dataset.xlsx",sheet_name="Dataset_5",nrows=10)

```

```

df["Weight"]=df["Weight"].apply(lambda x:"Heavy" if x>=140 else "Light")
df["Diameter"]=df["Diameter"].apply(lambda x:"Large" if x>=7 else "Small")
df["Sweetness"]=df["Sweetness"].apply(lambda x:"High" if x>=80 else "Low")
df["Fruit"]=df["Fruit"].replace({"Apple":"Yes","Orange":"No"})
concepts=df.iloc[:, :-1].values.tolist()
target=df.iloc[:, -1].tolist()
def candidate_elimination(concepts,target):
    S=concepts[0].copy()
    G=[["?" for i in range(len(S))] for j in range(len(S))]
    for i,h in enumerate(concepts):
        if target[i]=="Yes":
            for j in range(len(S)):
                if S[j]!=h[j]:
                    S[j]='?'
                    G[j][j]='?'
        else:
            for j in range(len(S)):
                if h[j]!=S[j]:
                    G[j][j]=S[j]
                else:
                    G[j][j]='?'
    G=[g for g in G if g!=["?"]*len(S)]
    print("Specific:",S)
    print("General:")
    for g in G:
        print(g)
candidate_elimination(concepts,target)

```

OUTPUT:

Specific: ['Heavy', 'Large', 'High']

General:

['Heavy', '?', '?']

['?', 'Large', '?']

['?', '?', 'High']

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